

Solution Requirements

Date	27 June 2026
Team ID	XXXXXX
Project Name	Rising waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>User authentication can be added to allow authorized users to access the flood prediction system securely. (Current version does not require login.)</i>	Medium
2	Authorization levels	<i>Administrators can manage the system, while users can enter weather data and view flood predictions.]</i>	Medium

3	External interfaces	The system accepts weather parameters through the Flask web application. Future versions can integrate real-time Weather APIs and IBM Cloud services.	High
4	Transactions processing	<i>The system receives user input, preprocesses the data, runs the trained XGBoost model, and displays the flood prediction instantly.</i>	High
5	Reporting	<i>Displays prediction results and model accuracy. Future versions may include dashboards and downloadable reports.</i>	Medium
6	Business rules, etc.	<i>Flood prediction is generated only after all required weather parameters are entered. Invalid or incomplete inputs are rejected with validation messages.</i>	High
7	Compliance to laws or regulations	<i>The application should follow general data privacy and security practices when handling user inputs and prediction data.</i>	Medium
8	Other	<i>The system should support future deployment on IBM Cloud and integration with notification services such as SMS and Email alerts.</i>	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	<i>The application should generate flood predictions quickly after receiving user input.</i>	<i>Prediction response time < 2 seconds</i>
2	Scalability	<i>The system should support multiple users and larger datasets after cloud deployment.</i>	<i>Supports concurrent users after IBM Cloud deployment</i>
3	Security & Data Privacy	<i>Protect user input through input validation and secure communication.</i>	<i>Validated inputs and HTTPS support after deployment</i>
4	Reliability & Availability	<i>The system should provide consistent prediction results with minimal downtime.</i>	<i>99% application availability</i>
5	Usability & Accessibility	<i>The web application should have a simple, responsive, and easy-to-use interface.</i>	<i>Responsive design compatible with desktop and mobile browsers</i>
6	Other	<i>The application should be modular, reusable, and easy to deploy on different platforms.</i>	<i>Runs on Windows, Linux, and IBM Cloud with minimal configuration</i>